Raisul Kabir News

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Research Interests

Building computing systems people can trust, from two directions: **designing systems that are trustworthy from the start** (privacy-preserving and verifiable federated learning, homomorphic encryption, zero-knowledge proofs, explainable AI) and **building tools that test systems until they break** (LLM-agent-driven property-based testing and fuzzing, program analysis, robustness under distribution shift).

Education

M.Sc. in Computer Science and Engineering Ongoing
BRAC University *Dhaka, Bangladesh*
- CGPA: 3.85/4.00, (expected CGPA: 3.96/4.00)

B.Sc. in Computer Science and Engineering 2019 – 2022
BRAC University *Dhaka, Bangladesh*
- CGPA: 3.91/4.00, graduated with Highest Distinction; Vice Chancellor's List (6 semesters) and Dean's List (2 semesters).

Publications

Journal Articles

[J1] An Advanced Data Fabric Architecture Leveraging Homomorphic Encryption and Federated Learning
Information Fusion, Elsevier, 2024 · Q1, Impact Factor 17.4 DOI arXiv
Secure medical image analysis using federated learning and homomorphic encryption, enabling privacy-preserving collaborative deep learning in healthcare. Received the BRAC University Quality Journal Publication Award (2025).

Conference Proceedings

[C1] Interpretable Deep Learning Approaches for Reliable GI Image Classification: A Study with the HyperKvasir Dataset
27th International Conference on Computer and Information Technology (ICCIT), IEEE, 2024 IEEE Xplore
Interpretable deep learning models for reliable gastrointestinal image classification.

Under Review

[R1] Why Cross-Dataset Transfer Fails in Human Activity Recognition (HAR): A Confound-Controlled Analysis of Classifier Degeneration
Submitted to Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT), 2026
Confound-controlled evaluation protocol identifying four methodological errors in prior HAR cross-dataset transfer studies, including a previously unreported PAMAP2 sensor-channel loading error. Introduces a label-free prediction-entropy diagnostic and a per-class CORAL alignment baseline for deployable wearable HAR systems.

Research in Progress

Verifiable Federated Aggregation as a Native Data Fabric Primitive (zk-FedMax) 2026
- Extending prior data fabric architecture with zero-knowledge proofs for selection-based federated aggregation (FedMax/FedMin/top-k), addressing the untrusted-server assumption left open in earlier work.
- Formalizing ZK-provable comparison/argmax operations for aggregator honesty, without revealing individual client updates.
- Targeting convergence guarantees under non-IID settings and empirical validation across multiple healthcare imaging datasets.

Quantifying Urban Heat Island Dynamics Using Spatio-Temporal Machine Learning Models 2026
Student research under my supervision
- Spatio-temporal forecasting of surface urban heat island intensity in Dhaka using multi-sensor Landsat/MODIS satellite data and ConvLSTM deep learning.
- Enables data-driven identification of heat-vulnerable urban zones and evaluation of vegetation-based mitigation strategies for rapidly urbanizing megacities.

Academic Appointments

Lecturer and Program Coordinator (Microprocessor Course) May 2024 – Present
Department of Computer Science and Engineering, BRAC University *Dhaka, Bangladesh*
- Teach undergraduate programming (Python, Java), microprocessors, algorithms, and data structures.
- Coordinate the microprocessor course: theory instruction, course content, and examinations.

Lecturer Jul 2023 – May 2024
Department of Information Technology and Management, Daffodil International University *Dhaka, Bangladesh*
- Taught Human-Computer Interaction and Artificial Intelligence; advised and mentored students, and evaluated labs, assignments, and examinations.

Adjunct Lecturer Sept 2023 – Apr 2024
Department of Computer Science and Engineering, BRAC University *Dhaka, Bangladesh*
- Conducted the Algorithms (CSE 221), Microprocessors (CSE 341), and Programming Language I (CSE 110) laboratory courses.

Student Tutor / Teaching Assistant Jul 2021 – Dec 2022
Department of Computer Science and Engineering, BRAC University *Dhaka, Bangladesh*
- Assisted in Microprocessors (CSE 341), Digital Logic Design (CSE 260), Algorithms (CSE 221), and Programming Language I (CSE 110).

Industry Experience

Junior Software Engineer Apr 2022 – Jul 2022
TechShoi *Dhaka, Bangladesh*
- Developed backend architecture using web frameworks and RESTful APIs for a large-scale enterprise system in Bangladesh.

Honors and Awards

Quality Journal Publication Award, BRAC University 2025
- For publishing a research article in a top 1% Q1 journal: *Information Fusion*, Elsevier (Impact Factor 17.4).

Open-Source Contribution: Upstream Fix in `tomlc17` 2026
- My agentic property-based fuzzer found a reproducible numeric-literal parsing bug in the open-source `tomlc17` parser; the maintainer confirmed and fixed it upstream (Issue #54, commit `e229ff5`).

Vice Chancellor's List and Dean's List, BRAC University 2019 – 2022
- Vice Chancellor's List: Fall 2019, Spring 2020, Fall 2020, Spring 2021, Summer 2021, Spring 2022. Dean's List: Summer 2019, Summer 2020.

Cultural Representation Award, BRAC University 2019
- For an outstanding performance in a cultural drama.

Selected Projects

Agentic Property-Based Fuzzing of tomlc17
LLM Agents, Property-Based Testing, Program Analysis github.com/RaisulKabir27/agentic-pbt-targeted-fuzzing-tomlc17
- Agentic property-based testing framework that iteratively generates and refines Hypothesis strategies for the tomlc17 TOML parser.
- Combines grammar-aware generation, sanitizer-based C harnesses, oracle-guided feedback, and automated failure triage.
- Discovered and reproduced a numeric-literal conformance bug, subsequently confirmed and fixed in the upstream repository.

VerifierProbe: Soundness Probe for Neural Network Verifiers
Neural Network Verification, Floating Point, PyTorch github.com/RaisulKabir27/nn-verifier-soundness-probe
- Tool that probes the CROWN bound-propagation engine of α,β-CROWN for floating-point soundness gaps, with a verifier adapter, a guided attacker, and a documented trace of the verifier's internal bounds.
- On a constructed network, float32 CROWN certified a margin of $+0.486$ while an input inside the region flipped the predicted class (float32 margin -24, float64 $+0.56$), reproducing the mechanism of Jia and Rinard (SAS 2021).
- Scoped the claim with controls: an ordinary network showed no violation, and the full α,β-CROWN pipeline caught the case in its attack stage.

Privacy-Preserving Medical Image Analysis

Federated Learning, Homomorphic Encryption github.com/RaisulKabir27/Advanced-Data-Fabric-Architecture

- Medical image analysis pipeline combining federated learning with homomorphic encryption.
- Implemented the Paillier cryptosystem in Python with modular arithmetic and Rabin–Miller primality testing; compared models on encrypted and unencrypted brain tumor images.

Secure Discord-Based Course Management Bot

Automation, Discord API, Backend Systems github.com/RaisulKabir27/Secure-Discord-Based-Course-Management-Bot

- Secure, automated Discord bot for course communication and coordination, built in my role as Program Coordinator.
- Role-based access control, automated announcements, and structured discussion channels, centralizing academic updates and administrative workflows.

DareDevil Line Follower Robot

Robotics, Embedded Systems, PID Control github.com/RaisulKabir27/DareDevil

- Autonomous Arduino-based line-following robot with infrared line sensors, DC motors, a motor driver, and a custom chassis.
- PID-based path tracking developed and tested in a Webots simulation of the robot and its track.

NUP Medical System

Django, Web Systems, PostgreSQL github.com/RaisulKabir27/NUPmedicals

- Hospital management system on Django's MVT architecture with modules for patient records, authentication, hospital workflows, and payments, backed by PostgreSQL.

Research Expertise

Federated and Distributed Learning	Designed and implemented decentralized frameworks for collaborative model training without centralized data sharing; applied to privacy-preserving medical image analysis (Q1 journal, *Information Fusion*).
Privacy-Preserving and Verifiable ML	Partially homomorphic encryption to secure sensitive data during distributed training and inference, addressing data-locality, security, and regulatory constraints; extending to zero-knowledge proofs so an untrusted aggregator can prove it is honest (ongoing).
Automated Testing and Program Analysis	Property-based testing, fuzzing, and program analysis to explore software behavior and isolate reproducible defects; built an LLM-agent-driven fuzzer whose finding was fixed upstream.
Explainable AI	Explainability techniques for transparency and clinical trust in deep learning; interpretability for GI image classification on HyperKvasir (IEEE conference publication).
Experimental Design and Benchmarking	Controlled experiments in centralized and federated settings; confound-controlled protocols exposing methodological errors in prior cross-dataset HAR studies; systematic benchmarking for peer-reviewed reporting.
Secure and Scalable System Design	Secure, automated AI-enabled systems with an emphasis on scalability and access control, demonstrated through a Discord-based course management platform.
AI for Societal Impact	AI for real-world problems such as healthcare analytics and urban heat risk; responsible AI practice (data privacy, transparency, fairness) through coursework and professional workshops.

Skills

TECHNICAL

Programming	Python, Java, C++, C#, Dart, Assembly; numerical computation and data processing
AI and Research Tools	PyTorch, NumPy, Pandas; model training, evaluation, and experimentation pipelines
Testing and Analysis	Hypothesis, property-based testing, fuzzing, C sanitizers
Web and Backend	Django, REST APIs, HTML, CSS; secure and automated academic platforms
Data and Toolchain	MySQL, MongoDB, PostgreSQL, Git, Linux, LaTeX, Android Studio

PROFESSIONAL

Independent Research	Planning, executing, and completing long-term research through peer-reviewed journal and conference publications
Scientific Communication	Explaining complex technical ideas through academic writing, classroom teaching, and research presentations
Leadership and Coordination	Program Coordinator and soccer team coach: organizing teams, managing responsibilities, guiding people to shared goals
Collaboration	Working with faculty members, co-authors, teaching assistants, and students in multi-role academic settings
Resilience and Time Management	Balancing research, teaching, coordination, and professional development under academic deadlines

Languages

English — Full professional proficiency

IELTS Overall	Listening	Reading	Writing	Speaking
7.0	7.5	7.5	6.5	6.5

Bangla — Native proficiency

Hindi — Limited working proficiency

Professional Training

Outcome-Based Education (OBE) and Bloom's Taxonomy
CLO–PLO mapping, assessment design, and curriculum alignment in higher education.
Instructor: Prof. Dr. Md. Mozahar Ali, Bangladesh Agricultural University

Online Teaching Tools and Digital Pedagogy
LMS usage, virtual assessments, and student engagement in online and hybrid classrooms.
Instructor: Mr. Narayan Ranjan Chakraborty, Associate Professor and Associate Head, CSE, Daffodil International University

AI Education and Applications
AI fundamentals, applied machine learning use cases, and ethical considerations in education.
Instructor: Mr. Amit Chakraborty, Assistant Professor, CSE, Daffodil International University

Leadership and Professional Skill Development
Academic leadership, communication skills, teamwork, and professional growth.
Instructor: Dr. Md. Sabur Khan, Chairman, Board of Trustees, Daffodil International University

Leadership and Activities

Soccer Team Coach, ITM Department, Daffodil International University Sept 2023 – Apr 2024

- Conducted training on possession play, pressing tactics, and counter-attacking strategies; the team finished runners-up in two inter-university tournaments.

Soccer, CSE Department, BRAC University Jul 2026

- Participated in an inter-university tournament with the department team.

Member, BRAC University Computer Club (BUCC) and Robotics Club (ROBU)

References

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